

**Exponential Expressions**Question Bank 3.03 **ANSWER KEY***Show your work. Circle your final answer.*

1. Two students evaluated  $\frac{(5^2 \cdot 2.8^0)^4}{5^7}$ . Patricia first wrote  $2.8^0$  as 1, then  $(5^2)^4 = 5^8$ , then  $\frac{5^8}{5^7} = 5$ . Kim first wrote  $(5^2)^4 \cdot (2.8^0)^4 = 5^8 \cdot 2.8^0$ , then  $5^8 \cdot 1$ , then  $\frac{5^8}{5^7} = 5$ .

- a. Name the law Patricia used first, and the law she used second.

*zero exponent, then power of a power*

- b. Name the law Kim used first.

*power of a product (the exponent 4 went to both factors)*

- c. Name the law both students used last, and give the value of the expression.

*quotient of powers; the value is 5*

**For questions 2 through 5, use the laws of exponents to evaluate. Show each law you use.**

2.  $\left(\frac{1.5^5}{1.5^3}\right)^2$

$(1.5^2)^2 = 1.5^4 = 5.0625$

3.  $\left[\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^1\right]^2$

$\left(1 \cdot \frac{4}{5}\right)^2 = \frac{16}{25}$

4.  $4.7^2 \cdot (4.7^1)^3$

$4.7^2 \cdot 4.7^3 = 4.7^5 = 2,293.45007$

5.  $\frac{9.9}{8.8^0} \cdot 9.9^3$

$\frac{9.9}{1} \cdot 9.9^3 = 9.9^4 = 9,605.9601$

**For questions 6 and 7, evaluate. Show each step.**

6.  $\frac{(2^2 \cdot 3.2^0)^6}{2^{10}}$

$\frac{2^{12}}{2^{10}} = 2^2 = 4$

$$7. \quad \frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$$\frac{(-4)^3}{1} = -64$$

**Equivalent or not.**

8. Select ALL of the expressions that are equivalent to  $\frac{0.2^3}{2^3} \cdot 0.1^2$ .

- ☒ A.  $0.1^5$
- ☒ B.  $0.1^3 \cdot 0.1^2$
- ☐ C.  $\frac{0.2^5}{0.2^3}$
- ☒ D.  $\left(\frac{0.2}{2}\right)^3 \cdot 0.1^2$
- ☐ E.  $0.2^3 \cdot 0.1^2$
- ☐ F.  $0.2^0 \cdot 0.1^2$

**A, B and D**

The expression is  $0.1^3 \cdot 0.1^2 = 0.1^5 = 0.00001$ .

9. Which expression is equivalent to  $\left(\frac{3^5}{3^2}\right)^2$ ?

- ☐ A.  $3^5$
- ☒ B.  $3^6$
- ☐ C.  $3^{14}$
- ☐ D.  $3^7$

**B**

$3^3$  squared is  $3^6 = 729$ .

**Working backwards.**

10. Find the value of  $n$  that makes each statement true.

a.  $\left(\frac{2^7}{2^n}\right)^2 = 2^6$

**$n = 4$**

Inside must be  $2^3$ .

b.  $\frac{5^n \cdot 5^0}{5^3} = 25$

**$n = 5$**

$5^{n-3} = 5^2$ .

11. Ruth says that  $\frac{(-15)^4}{1.5^4}$  can be rewritten as  $\left(\frac{-15}{1.5}\right)^4$  before evaluating. Is she right? Name the law, and give the value.

*Yes — power of a quotient, read from right to left;  $(-10)^4 = 10,000$  .*